Sample Registration System is adopted at both national and state levels in India to collect the following information:

National Level

(a) Infant mortality

(b) Age specific mortality rates in rural areas

(c) Sampling variability of vital rates

State Level

(a) Differences in birth rates with respect to education, religion, parity

(b) Sex ratio

(a) Seasonality in birth and death rates

(v) Analytical Method

It is generally not possible to conduct *ad hoc* surveys to assess the population at any specific period between two consecutive census years. Population estimates of any vital event at a given time can be obtained without *ad hoc* surveys applying analytical methods which use mathematical formulae.

Calculation of Vital Rates

Generally, rate of a vital event is calculated using the formula

$$\textbf{Rate of a vital event} = \frac{Number\ of\ occurrences\ of\ the\ event\ during\ the\ specified\ period}{Size\ of\ the\ population\ exposed\ to\ the\ risk\ of\ the\ event} \times 1000$$

Rates of vital events are usually expressed '*per thousand*'.

8.1.3 Mortality and Its Measurements

Mortality refers to the deaths, which occur in the population/community/region due to sickness, accidents, etc.

Several rates are used for measuring mortality. We will discuss the following primary mortality rates.

(i) Crude Death Rate

(ii) Specific Death Rate

(iii) Infant Mortality Rate

(i) Crude Death Rate

Crude Death Rate (*CDR*) is the simplest type of death rate, which relates the number of deaths in a specific community or region to the population size of the community in a given

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